

Uncertainty in AI

- Uncertainty refers to situations where the outcome or state of information is **not deterministic**.
- Uncertainty in Artificial Intelligence (AI) refers to the inability of models to make fully confident predictions due to incomplete, ambiguous, or **noisy data**.
- AI systems must account for uncertainty to make accurate and reliable decisions, especially in dynamic environments where information is inconsistent or evolving.
- AI does not always give **100% correct** or confident outputs. Sometimes the data is unclear, so the model becomes unsure.

Why Uncertainty Happens?

Uncertainty happens because:

1. Data is unclear

Example: A blurry photo.

2. Data is missing

Example: A few values are not available in a dataset.

3. Data is confusing

Example: Two objects look very similar.

4. The model has not learned enough

Example: The model never saw that type of example before.

Techniques for Addressing Uncertainty in AI

- AI systems use different methods to **measure**, **reduce**, and **manage** uncertainty so they can make safer and more accurate decisions.

1. Bayesian Belief Networks

Bayesian Belief Networks

Definition:

BBNs are graphical models that represent probabilistic relationships among variables and their conditional dependencies using a directed acyclic graph(DAG).

- It is used to handle uncertainty and make predictions or decisions based on probabilities.
- Bayesian Belief Networks are valuable tools for understanding and solving problems involving uncertain events. They are also known as Bayes networks, belief networks, decision networks, or Bayesian models.

Bayesian Belief Networks

Basic Structure of Bayesian Networks

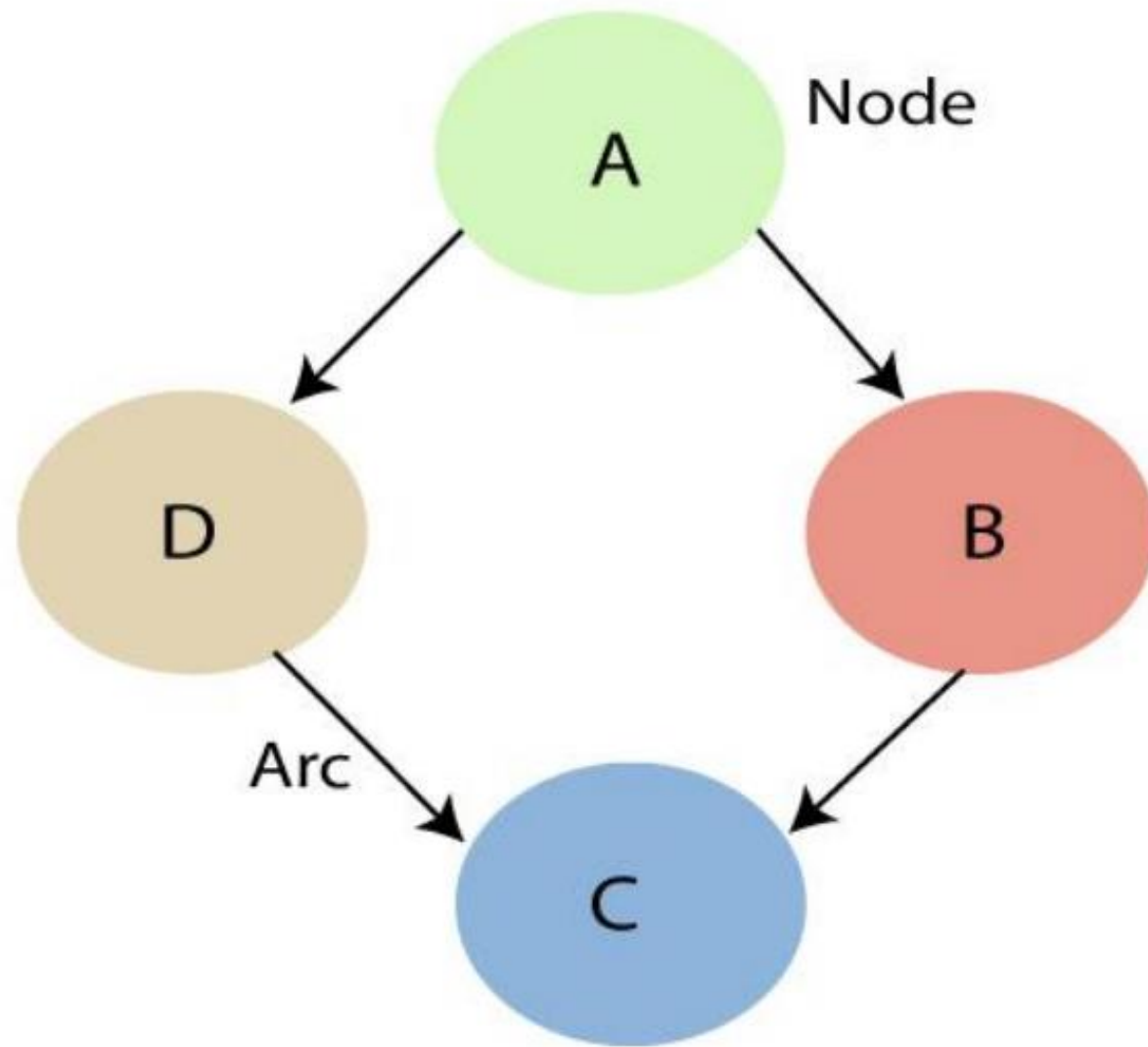
- **Nodes:** Each node represents a random variable, which can be discrete or continuous.
- **Edges:** Directed edges (arrows) between nodes represent conditional dependencies.
- **Conditional Probabilities:** Each node's probability depends on its parent nodes, expressed as $P(\text{Variable} \mid \text{Parent})$.

For example, if node A influences node B, there would be a directed edge from A to B, indicating that B is conditionally dependent on A.

Bayesian Belief Networks

Bayesian Network consists of two parts:

- Directed Acyclic Graph
- Table of conditional probabilities



Bayesian Belief Networks

- A Bayesian network graph is made up of **nodes and Arcs** (directed links), where:
- Each node corresponds to the **random variables**, and a variable can be **continuous or discrete**.
- **Arc or directed arrows** represent the **causal relationship or conditional probabilities between random variables**. These directed links or arrows connect the pair of nodes in the graph. These links represent that **one node directly influence the other node**, and if there is no directed link that means that nodes are independent with each other
- In the above diagram, A, B, C, and D are random variables represented by the nodes of the network graph.
- If we are considering node B, which is connected with node A by a directed arrow, then node A is called the parent of Node B.
- Node C is independent of node A.

Note: The Bayesian network graph does not contain any cyclic graph. Hence, it is known as a directed acyclic graph or DAG

Bayesian Network



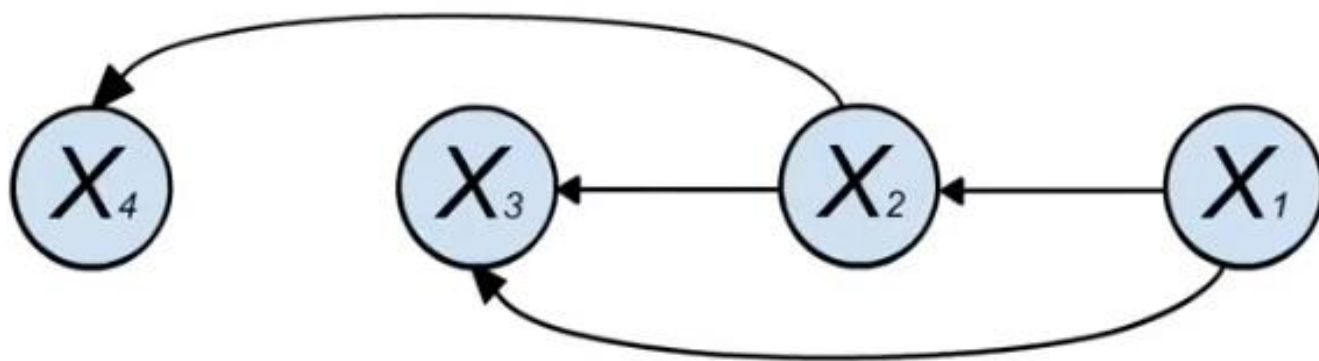
Conditional Probability Table

Condition	Probability
Traffic = High Rain = No	70%
Accident = Yes Traffic = High	20%
Accident = Yes Traffic = Low	40%

Basic Probabilistic Notations

- Probability notation is an efficient way of writing the probability of events happening or not happening.

Probability Notation	Description
$P(A)$	The probability of event A occurring
$P(A')$	The probability of event A not occurring
$P(A \cap B)$	The probability of both A and B occurring at the same time
$P(A \cup B)$	The probability of either A or B occurring
$P(A \cap B')$	The probability of A occurring but not B
$P(A' \cup B)$	The probability of either A not occurring or B occurring



$$P(X_4 | X_2) \quad P(X_3 | X_1, X_2) \quad P(X_2 | X_1) \quad P(X_1)$$

$$P(X_1, X_2, X_3, X_4) = P(X_4 | X_2)P(X_3 | X_1, X_2)P(X_2 | X_1)P(X_1)$$

Conditional Probability (CP)

- **Conditional Probability** is the probability of an event occurring **given that another event has already occurred**.
- The conditional probability of an event A given B is written as:
- $P(A | B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$

Where:

- $P(A | B)$ = Probability of **A occurring given B**
- $P(A \cap B)$ = Probability that **both A and B occur**
- $P(B)$ = Probability that **B occurs**

Conditional Probability (CP)

- Useful for understanding cause-and-effect relationships; i.e., helps predict outcomes based on known information.

Conditional Probability in Real Life

- **Medical Diagnosis:** “Given a positive test result, what is the chance the patient has the disease?”
- **Spam Email Detection:** “Given that an email contains the word ‘free’, what is the probability it’s spam?”
- **Weather Forecast:** “Given that it’s cloudy, what is the probability of rain?”
- These are all **conditional probabilities**.

Conditional Probability (CP)

Connection to Bayes' Theorem

- Bayes' Theorem is based on conditional probability:

- $$P(A | B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$ the probability of event A occurring, given event B has occurred
- $P(B|A)$ the probability of event B occurring, given event A has occurred
- $P(A)$ the probability of event A
- $P(B)$ the probability of event B

Conditional Probability

- **Example 1:** In a class, 30% of the students failed in Physics, 25% failed in Mathematics and 15% failed in both Physics and Mathematics. If a student is selected at random failed in Mathematics, find the probability that he failed in Physics also.

Let A be the event "failed in Physics" and B be the event "failed in Mathematics". We want to find $P\left(\frac{A}{B}\right)$.

It is given that $P(A) = \frac{30}{100}$ and $P(B) = \frac{25}{100}$

Also $P(A \cap B) = \frac{15}{100}$

Therefore $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{15}{100}}{\frac{25}{100}} = \frac{15}{25} = \frac{3}{5}$

Using conditional probability formula

Joint Probability

- The probability of two or more events occurring simultaneously.
- **Calculation** $P(A \cap B)$, $P(A, B)$
- **Variables involved** Two or more events

This means:

- Event **A** happens
- AND event **B** happens
Together

Formula for Joint Probability

- **1. For Independent Events**

- When events A and B are independent, meaning that the occurrence of one event does not impact the other, we use the multiplication rule:

- $P(A \cap B) = P(A) \times P(B)$

-

Here, $P(A)$ is the probability of occurrence of event A, $P(B)$ is the probability of occurrence of event B, and $P(A \cap B)$ is the joint probability of events A and B.

Formula for Joint Probability

2. For Dependent Events

Events are often dependent on each other, meaning that one event's occurrence influences the likelihood of the other. Here, we employ a modified formula:

$$P(A \cap B) = P(A) \times P(B|A)$$

Here, $P(A)$ is the probability of occurrence of event A, $P(B|A)$ is the conditional probability of occurrence of event B when event A has already occurred, and $P(A \cap B)$ is the joint probability of events A and B.

General Joint Probability Formula

- For variables A, B, C:
- $P(A,B,C)=P(A)\cdot P(B|A)\cdot P(C|A,B)$
- **P(A)** This is the probability of the **first event** happening.
- **P(B | A)** This means: **Probability that B happens *given that A already happened*.**
- **P(C | A, B)** This means: **Probability that C happens when BOTH A and B have already happened.**

Examples of Joint Probability

- **Example 1: Independent Events**

- Suppose you are running an e-commerce platform, and you want to find the probability of a customer purchasing a red shirt (event A) and a blue hat (event B) independently. Find out the Joint Probability where
- $P(A)$: The probability of a customer buying a red shirt is 0.3.
- $P(B)$: The probability of a customer purchasing a blue hat is 0.2.
- **Solution:**
- $P(A \cap B) = P(A) \times P(B)$
- $P(A \cap B) = P(\text{customer buying a red shirt}) \times P(\text{customer buying a blue hat})$
- $P(A \cap B) = 0.3 \times 0.2$
- **$P(A \cap B) = 0.06$**

Examples of Joint Probability

Example 2: Dependent Events

- Imagine you are in the insurance business, and you want to determine the probability of a customer filing a claim (event A) and receiving a payout (event B), given that a claim was filed. Find out the Joint Probability where
- $P(A)$: The probability of a customer filing a claim is 0.1.
- $P(B | A)$: The probability of a customer receiving a payout given that a claim was filed is 0.8.
- **Solution:**
- $P(A \cap B) = P(A) \times P(B|A)$
- $P(A \cap B) = P(\text{customer filing a claim}) \times P(\text{customer receiving a payout given that a claim was filed})$
- $P(A \cap B) = 0.1 \times 0.8$
- **$P(A \cap B) = 0.08$**

BBN Example

Problem Statement:

Consider a medical diagnosis scenario where a patient shows symptoms that may indicate either a **Cold (C)** or the **Flu (F)**. The symptoms are **Fever (F)**, **Runny Nose (R)**, and **Cough (Cg)**. Based on medical data, the following probabilities are known:

- $P(F)$:Probability of having Flu = **0.1**
- $P(C)$:Probability of having Cold = **0.2**
- $P(F | F)$:Probability of Fever given Flu = **0.8**
- $P(R | F)$:Probability of Runny Nose given Flu = **0.6**
- $P(Cg | F)$:Probability of Cough given Flu = **0.4**
- $P(F | C)$:Probability of Fever given Cold = **0.6**
- $P(R | C)$:Probability of Runny Nose given Cold = **0.4**
- $P(Cg | C)$:Probability of Cough given Cold = **0.3**

Bayesian Belief Network Construction

- The Bayesian Belief Network (BBN) for this medical scenario is structured as follows:

1. Nodes

- **Disease (D)** with states: {Cold (C), Flu (F)}
- **Symptoms (S)** with three nodes:
 - Fever (F)
 - Runny Nose (R)
 - Cough (Cg)

Bayesian Belief Network Construction

2. Edges

- Directed edges go **from Disease (D) to each Symptom node**, showing that the disease causes the symptoms.

Bayesian Belief Network Construction

3. Conditional Probabilities

- **Disease priors:**

- $P(D = F) = 0.1$
- $P(D = C) = 0.2$

- **Symptom likelihoods given Flu:**

- $P(F | F) = 0.8$
- $P(R | F) = 0.6$
- $P(Cg | F) = 0.4$

- **Symptom likelihoods given Cold:**

- $P(F | C) = 0.6$
- $P(R | C) = 0.4$
- $P(Cg | C) = 0.3$

Given that a patient has Fever (F), what is the probability that they have the Flu (F)?

- To find this, we apply **Bayes' Theorem**:
- $$P(Flu | Fever) = \frac{P(Flu) \cdot P(Fever|Flu)}{P(Flu) \cdot P(Fever|Flu) + P(Cold) \cdot P(Fever|Cold)}$$
- Substitute values:
- Numerator = $0.1 \times 0.8 = 0.08$
- Denominator = $0.1 \times 0.8 + 0.2 \times 0.6 = 0.08 + 0.12 = 0.20$
- So,
- $$P(Flu | Fever) = \frac{0.08}{0.20} = 0.4$$

Example 2

- A patient presents with Runny Nose (R). Calculate the probability that they have a Cold (C).

$$P(C|\text{Runny Nose}) = \frac{P(C) \cdot P(\text{Runny Nose}|C)}{P(\text{Runny Nose})}$$

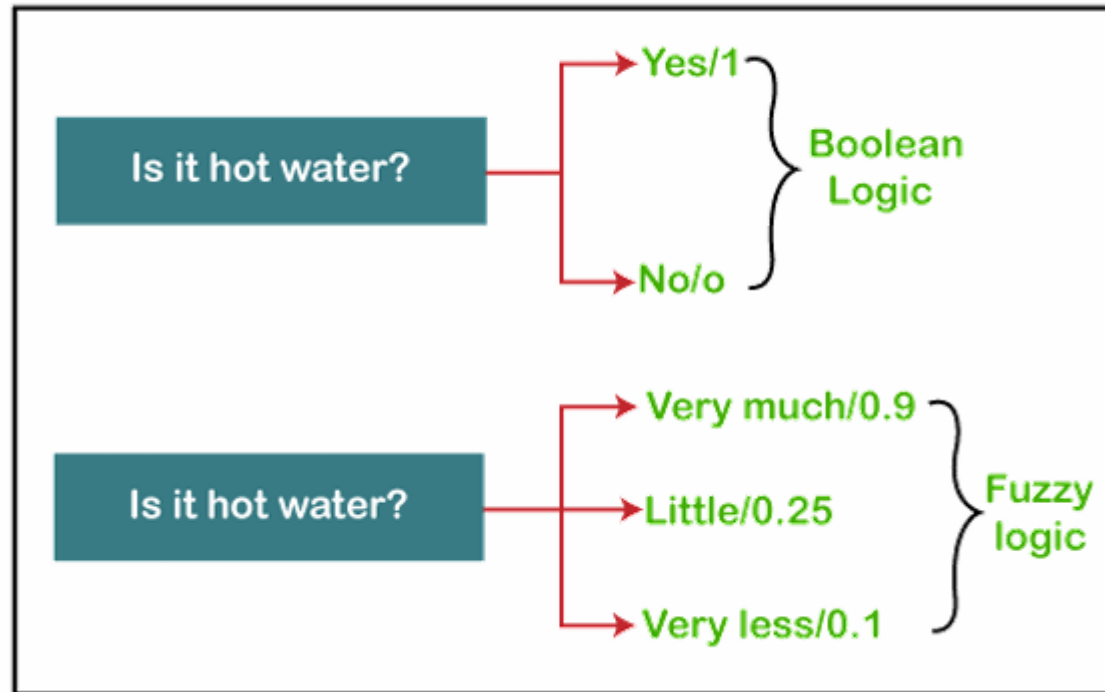
First, calculate $P(\text{Runny Nose})$:

$$P(\text{Runny Nose}) = P(F) \cdot P(\text{Runny Nose}|F) + P(C) \cdot P(\text{Runny Nose}|C)$$

Fuzzy Logic

- Fuzzy Logic is a method designed to handle **uncertainty** and imprecision in situations where things are not clear.
- Fuzziness occurs when the boundary of a piece of information is not clear
- Example of Fuzzy Logic as comparing to Boolean Logic

Fuzzy Logic



Fuzzy Logic

- **Example:** Words like **young, tall, good or high** are fuzzy.
- There is no **single quantitative** value which defines the term young.
- For some people, **age 25 is young**, and for others, **age 35** is young. The concept young has no clean boundary.
- Age **35** has some possibility of being **young** and usually depends on the context in which it is being considered.
- Fuzzy set theory is an extension of **classical set** theory where elements have **degree of membership**. To overcome this difficulty in describing vague concepts, membership value is introduced.

Classical and Fuzzy Set Theory

Set

- A set is a term, which is a collection of unordered or ordered elements. Following are the various examples of a set:
 1. A set of all-natural numbers
 2. A set of students in a class.
 3. A set of all cities in a state.
 4. A set of upper-case letters of the alphabet.

Types of Set

1. Empty Set (Null Set)

- A set with **no elements**.

Example:

$$A = \{\}$$

$$A = \emptyset$$

2. Finite Set

- A set with **countable** elements.

Example:

$$B = \{1, 2, 3, 4\}$$

3. Infinite Set

- A set with **uncountable** or **endless** elements.

Example:

$$C = \{1, 2, 3, 4, \dots\}$$

Types of Set

4. Equal Sets

- Two sets that have **exactly the same elements**.
Example:
 $\{1, 2, 3\} = \{3, 2, 1\}$

5. Equivalent Sets

- Sets that have the **same number of elements**, even if they are different.
Example:
 $A = \{a, b, c\}$
 $B = \{1, 2, 3\} \rightarrow$ Both have 3 elements

- 6. Subset**

- A set whose **all elements** are inside another set.
Example:
 $A = \{1, 2\}$
 $B = \{1, 2, 3\}$
 $A \subseteq B$

Types of Set

7. Proper Subset

- A set that is a subset **but not equal**.
Example:
 $A = \{1, 2\}$ is a proper subset of $B = \{1, 2, 3\}$

8. Universal Set

- A set that contains **all possible elements** in a given context.
Usually denoted by **U**.
Example:
 $U = \{1, 2, 3, 4, 5\}$

9. Power Set

- A set of **all subsets** of a set.
Example:
 $A = \{1, 2\}$
Power set = $\{\{\}, \{1\}, \{2\}, \{1, 2\}\}$

10. Singleton Set

- A set with **only one element**.
Example:
 $A = \{5\}$

Fuzzy set

- A fuzzy set is a collection of values which exist **between 0 and 1**.
- Fuzzy sets are denoted or represented by **the tilde (~) character**.
- In the fuzzy set, the **partial membership** also exists.
- This theory released as an extension of classical set theory.

This theory is denoted mathematically as **A fuzzy set ($\tilde{A}$) is a pair of U and M , where U is the Universe of discourse and M is the membership function** which takes on values in the interval **[0, 1]**. The **universe of discourse (U)** is also denoted **by Ω or X** .

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) | x \in X\}$$

Operations on Fuzzy Set

- Given $\tilde{A}$ and B are the two fuzzy sets, and X be the universe of discourse with the following respective member functions:

$$\mu_{\tilde{A}}(x) \text{ and } \mu_{\tilde{B}}(x)$$

- The operations of Fuzzy set are as follows:
- 1. Union Operation:** The union operation of a fuzzy set is defined by:

$$\mu_{A \cup B}(x) = \max (\mu_A(x), \mu_B(x))$$

2. Intersection Operation: The intersection operation of fuzzy set is defined by:

$$\mu_{A \cap B}(x) = \min (\mu_A(x), \mu_B(x))$$

3. Complement Operation: The complement operation of fuzzy set is defined by:

$$\mu_{\bar{A}}(x) = 1 - \mu_A(x),$$

1. Fuzzy Union ($A \cup B$)

Combines two fuzzy sets by taking the **maximum** membership value.

Let fuzzy sets A and B be:

x	$\mu_A(x)$	$\mu_B(x)$
1	0.2	0.5
2	0.7	0.4
3	0.1	0.8

Now compute union:

$$A \cup B = \max(\mu_A, \mu_B)$$

x	$\mu_A(x)$	$\mu_B(x)$	$\mu_{A \cup B}(x)$
1	0.2	0.5	0.5
2	0.7	0.4	0.7
3	0.1	0.8	0.8

Fuzzy Intersection ($A \cap B$)

x	$\mu_A(x)$	$\mu_B(x)$	$\mu_{A \cap B}(x)$
1	0.2	0.5	0.2
2	0.7	0.4	0.4
3	0.1	0.8	0.1

Fuzzy Complement (A^c)

For fuzzy set A:

x	$\mu_A(x)$	$\mu_{A^c}(x)$
1	0.2	0.8
2	0.7	0.3
3	0.1	0.9

Applications of Fuzzy Logic

1. It is used in **Businesses** for decision-making support system.
2. It is used in **Automotive systems** for controlling the traffic and speed, and for improving the efficiency of automatic transmissions. **Automotive systems** also use the shift scheduling method for automatic transmissions.
3. It is also widely used in the **Pattern Recognition and Classification** in the form of Fuzzy logic-based recognition and handwriting recognition. It is also used in the searching of fuzzy images.
4. Fuzzy logic systems also used in **Securities**.
5. It is also used in **microwave oven** for setting the power and cooking strategy.
6. This technique is also used in the area of **modern control systems** such as expert systems.
7. **Finance** is also another application where this concept is used for predicting the stock market, and for managing the funds.
8. It is also used for controlling the brakes.
9. It is also used in the **industries of manufacturing** for the optimization of milk and cheese production.
10. It is also used in the vacuum cleaners, and the timings of washing machines.
11. It is also used in heaters, air conditioners, and humidifiers.

Advantages of Fuzzy Logic

1. Fuzzy logic relies on logical sets and reasoning which can be easily understood by any user.
2. Fuzzy logic is easy to understand and hence fuzzy logic is user-friendly.
3. Many complex problems cannot be solved by other systems which are easily solved by fuzzy logic.
4. Fuzzy logic can give accurate outputs with imprecise data or inaccurate.
5. Fuzzy logic is widely used for commercial and practical purposes.

Disadvantages of Fuzzy Logic

1. The run time of fuzzy logic systems is slow and takes a long time to produce outputs.
2. Users can understand it easily if they are simple.
3. The possibilities produced by the fuzzy logic system are not always accurate.
4. Fuzzy logics are not suitable for those problems that require high accuracy.
5. The systems of a Fuzzy logic need a lot of testing for verification and validation.

Classical Set Theory	Fuzzy Set Theory
1. This theory is a class of those sets having sharp boundaries.	1. This theory is a class of those sets having un-sharp boundaries.
2. This set theory is defined by exact boundaries only 0 and 1.	2. This set theory is defined by ambiguous boundaries.
3. In this theory, there is no uncertainty about the boundary's location of a set.	3. In this theory, there always exists uncertainty about the boundary's location of a set.
4. This theory is widely used in the design of digital systems.	4. It is mainly used for fuzzy controllers.

Why Fuzzy Logic?

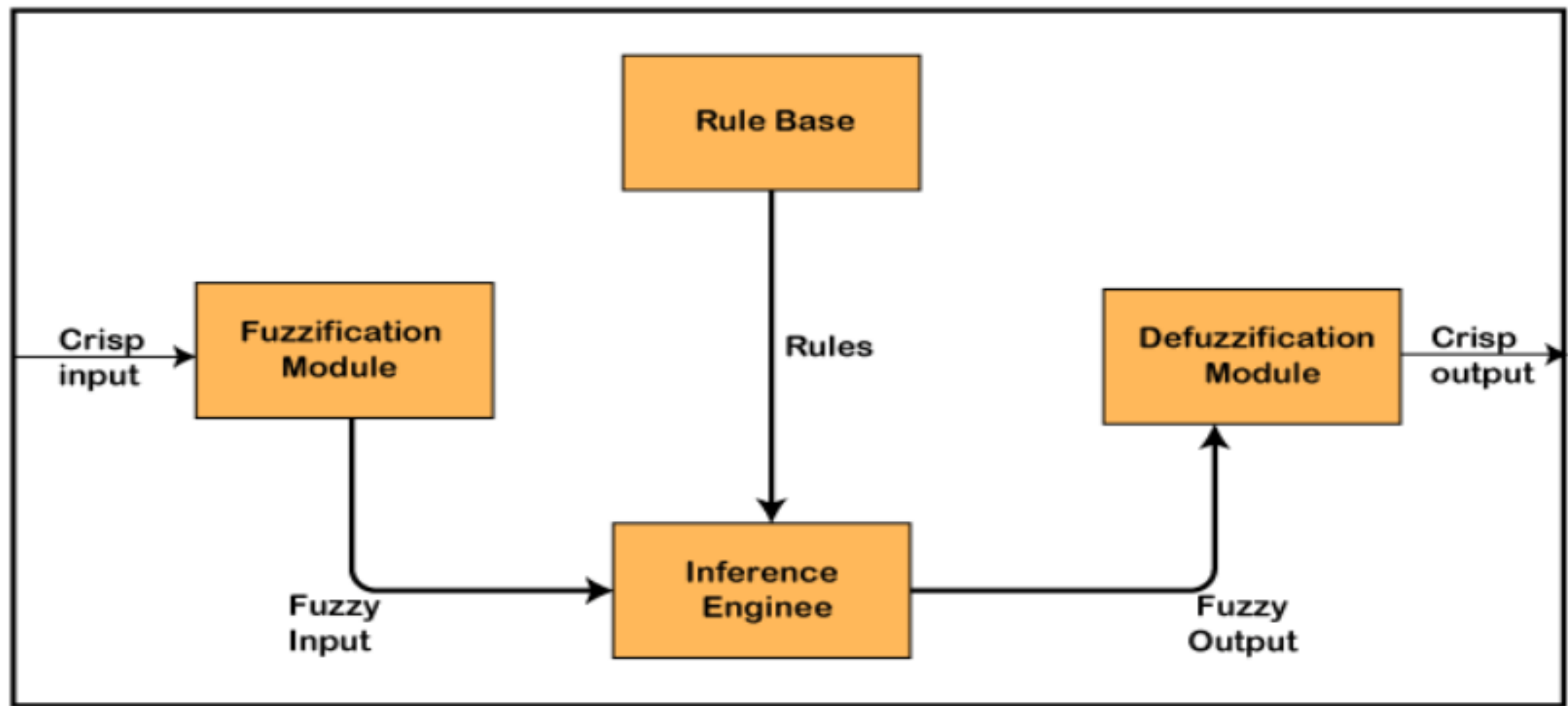
- Fuzzy logic is integral in artificial intelligence for its ability to mimic the nuances of human reasoning, dealing adeptly with uncertainty and imprecision.
- Unlike traditional binary logic, fuzzy logic accommodates the complexities of real-world situations by allowing for degrees of truth rather than rigid true/false dichotomies.
- This flexibility enables AI systems to make more practical and realistic decisions in scenarios where information is ambiguous or incomplete. By embracing the variability and subtlety of human thought processes, fuzzy logic significantly enhances the adaptability and effectiveness of AI applications, making them more aligned with the unpredictable nature of real-world environments.

Fuzzy Logic Architecture

- In the architecture of the Fuzzy Logic system, each component plays an important role.

Components of a Fuzzy Logic Controller

- The following are the components of a fuzzy logic system/controller:
 1. Knowledge Base
 2. Fuzzification Module
 3. Inference Engine
 4. Defuzzification Module



Fuzzy Rule Base:

- The core of the **fuzzy logic system**, the rule base consists of a collection of fuzzy **IF-THEN** rules.
- These rules use **natural language terms** (like "high," "low," "warm") to describe **conditions and actions**.
- IF Temperature is Warm THEN Fan Speed is Medium.
- IF Temperature is Cold THEN Fan Speed is Low.

Fuzzification

- Fuzzification is **a module or component** for transforming the system inputs, i.e., it converts the **crisp number(numerical) inputs into fuzzy steps**. The crisp numbers are those inputs which are measured **by the sensors** and then fuzzification passed them into the control systems for further processing.
- **Crisp value:** 'Crip' means precise, It deal with the values that have strict boundaries, i.e. True and False

- This component divides the input signals into following **five states** in any Fuzzy logic system
- Large Positive (LP)
- Medium Positive (MP)
- Small (S)
- Medium Negative (MN)
- Large negative (LN)Logic system:

Inference Engine

- The Inference engine is the main component of the Fuzzy Logic System. It acts as a kernel of any FLC. Basically, it simulates human decisions by performing approximate reasoning.
- It evaluates the fuzzified input data and applies the relevant rules from the rule base. It find which rules are most appropriate and then combines the results to generate a control action.

Defuzzification

- Once the inference engine processes the rules and generates fuzzy outputs, it converts these fuzzy results back into a precise, crisp value. This final step ensures that the system can take clear actions based on the fuzzy logic outputs.